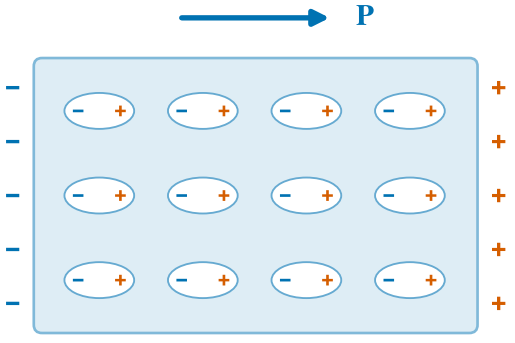
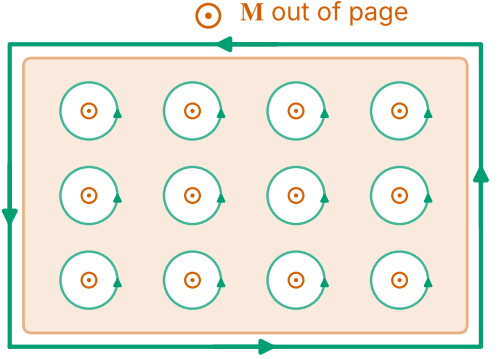
Uniformly polarized slab



interior $\rho_b = -\nabla \cdot \mathbf{P} = 0$

surfaces $q_{b,\,s} = \mathbf{P} \cdot \mathbf{n}$: − left, + right

Uniformly magnetized slab (cross-section)



interior $\mathbf{J}_M = \nabla \times \mathbf{M} = 0$

boundary $\mathbf{K}_M = \mathbf{M} \times \mathbf{n}$: circulates